

Automatic Diagnosis of Diabetic Retinopathy via Custom CNN Architecture and Transfer Learning

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Introduction

Diabetic Retinopathy (DR) is a leading cause of preventable blindness. This work develops a binary DR/No-DR classifier from fundus photographs across two competition tracks: a *custom* CNN trained from scratch (CUSTOM) and a *fine-tuned* pretrained backbone (FINE-TUNING). Performance is measured by AUC on a held-out 1 000-image test set evaluated via Codabench.

Dataset & Preprocessing

The dataset contains 2 000 training / 500 validation / 1 000 test fundus images with a severe class imbalance (73% No-DR, 27% DR). Right-eye images are horizontally mirrored to unify anatomical orientation.

Pipeline. A threshold-based `CropByEye` transform crops the visible retinal disc, removing black borders. Images are rescaled to 256 px; training augmentations add random horizontal flip, rotation ($\pm 15^\circ$), colour jitter, and a random 224 px crop. Validation uses a deterministic centre crop. All images are ImageNet-normalised ($\mu=[0.485, 0.456, 0.406]$; $\sigma=[0.229, 0.224, 0.225]$).

Custom Model (CUSTOM)

Baseline: CustomNet

A 4-block CNN with 5×5 no-padding convolutions ($3 \rightarrow 6 \rightarrow 16 \rightarrow 32 \rightarrow 64$), BatchNorm and Max-Pool per block, flattened to 6 400-d, and head $6400 \rightarrow 120 \rightarrow 84 \rightarrow \text{Dropout}(0.5) \rightarrow 1$. SGD + StepLR; AUC = 0.570.

CustomNetV2

To improve generalisation on the small training set:

- **5 blocks, 3×3 same-padding** convolutions ($3 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 256$), reducing spatial size with MaxPool per block.
- **Global Average Pooling** replaces the 6 400-d flatten, giving translation invariance and a compact 256-d representation.
- **Head:** $\text{Dropout}(0.2) \rightarrow \text{Linear}(256, 128) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.4) \rightarrow \text{Linear}(128, 1)$.
- **Optimiser:** AdamW ($\text{lr}=1\text{e-}3$, $\text{wd}=5\text{e-}3$), CosineAnnealingLR ($T_{\text{max}}=50$), 50 epochs, patience=10.

Adding `WeightedRandomSampler` raises public AUC from 0.744 to **0.766** (+34% over baseline).

Fine-tuning Model (FINE-TUNING)

Backbone: EfficientNet-B0

ResNet18 [2] yielded 0.698 AUC. We switched to **EfficientNet-B0** [1] (ImageNet top-1 77.1%), replacing its classifier with $\text{Dropout}(0.3) \rightarrow \text{Linear}(1280, 1)$. Unfreezing only `features[6-8]` plus the new head (3.16 M of 4.01 M parameters), with AdamW [3] ($\text{lr}=3\text{e-}4$, $\text{wd}=1\text{e-}2$) and CosineAnnealingLR ($T_{\text{max}}=30$) for 30 epochs, raised AUC to 0.751.

Two-Stage Progressive Unfreezing

Fine-tuning all layers on 2 000 images risks overfitting. *Stage 1* trains only deep blocks (`features[6-8]`) + head. *Stage 2* additionally unfreezes `features[5]` at $\text{lr}=3\text{e-}5$, triggered only if Stage 1 TTA AUC ≥ 0.752 ; Stage 1 weights are restored if Stage 2 regresses. Despite this, overfitting persists: train AUC reaches 0.957 at the best val checkpoint with a 0.178 train-val gap (Fig. 1).

Training Optimizations

Class-imbalance handling. `BCEWithLogitsLoss` with `pos_weight  $\approx 2.76$` addresses imbalance at the gradient level. `WeightedRandomSampler` oversamples the minority class at the mini-batch level (+0.022 AUC).

Test-Time Augmentation (TTA). Scores are averaged over the original and horizontally-flipped image (+0.008–0.012 AUC, applied at inference only).

Label Smoothing. For EfficientNet only ($\epsilon=0.05$), hard targets become soft $\{0.025, 0.975\}$, penalising overconfident logits without damaging the limited training signal on a small dataset [4].

Ensemble. A grid search over $w \in \{0.10, \dots, 0.30\}$ on validation selects the optimal blend of CustomNetV2 and EfficientNet scores. A val-test gap of ≈ 0.02 – 0.03 AUC is consistently observed (val 0.796 $\rightarrow$ public 0.770).

Ablations & Negative Results

- **CLAHE preprocessing:** histogram equalisation before `CropByEye` broke its intensity threshold, dropping custom AUC by 4.6% ($0.570 \rightarrow 0.544$).
- **Cache pipeline:** pre-computing fixed 224 px crops removed per-epoch `RandomCrop` spatial jitter — a key regulariser on 2 000 images — costing -0.028 AUC despite a $5 \times$ speed-up.
- **Scale jitter (TVRandomResizedCrop):** no consistent gain (-0.004 vs best).
- **BenGraham preprocessing:** local contrast normalisation caused catastrophic collapse (custom 0.541, ft 0.558) due to a dtype bug: the transform assumed float $[0, 1]$ input but received uint8 $[0, 255]$ from `CropByEye`. `clip(uint8, 0, 1)` mapped all pixels to $\{0, 1\}$; the contrast formula produced uniform grey images, making every training sample indistinguishable.

Individual Contributions

Cristina [Surname]: Baseline exploration in `first_attempt.ipynb` — class-imbalance analysis, CustomNet design, initial data pipeline, and first Codabench submission.

Maria Muñoz: All subsequent work (experiments B–H) — CustomNetV2 design, EfficientNet-B0 two-stage fine-tuning, `WeightedRandomSampler`, TTA, ensemble grid search, label smoothing, scale-jitter and cache ablations, and all notebook documentation.

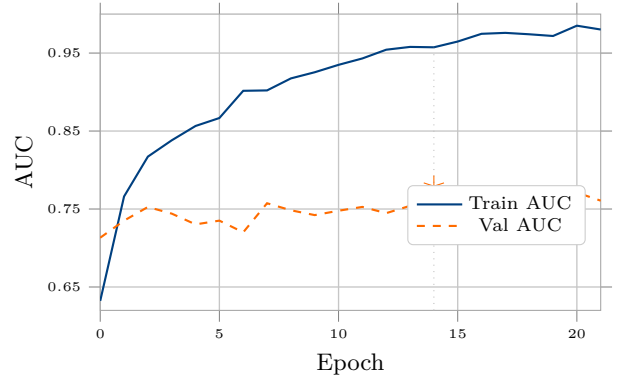
Table 1: AUC results for all submitted experiments (*pub* = public leaderboard, *priv* = private). **Bold:** best result. *Italic / red rows:* discarded experiment.

Notebook	Key change(s)	Custom AUC		FT AUC	
		pub / priv		pub / priv	
first_attempt	CustomNet (baseline) + ResNet18 fine-tuning	0.570 / 0.597		0.698 / 0.708	
attempt_with_b	CustomNet: Adam optimiser + CosineAnnealingLR	0.558 / 0.566		0.692 / 0.693	
attempt_with_a_b	<i>CLAHE preprocessing — broke CropByEye (−4.6%)</i>	<i>0.544 / 0.562</i>		<i>0.711 / 0.692</i>	
attempt_with_c	FT: switch to EfficientNet-B0 + AdamW	0.561 / 0.579		0.751 / 0.734	
second_attempt	CustomNetV2; 2-stage progressive FT; TTA; Ensemble	0.744 / 0.731		0.742 / 0.727	
second_attempt_tuned	+ WeightedRandomSampler (balanced mini-batches)	0.766 / 0.743		0.770 / 0.744	
s.a.t._cache	<i>Cache pipeline — removes RandomCrop jitter (−0.028)</i>	<i>0.736 / 0.730</i>		<i>0.742 / 0.728</i>	
third_attempt	Scale jitter (ResizedCrop); label smoothing; $T_{\max}=25$	0.762 / 0.742		0.766 / 0.743	
fourth_attempt	<i>BenGraham — uint8 dtype bug, all images grey (−22%)</i>	<i>0.541 / 0.557</i>		<i>0.558 / 0.533</i>	

Table 2: Architecture and training comparison.

Feature	CustomNet	CustomNetV2
Conv blocks	4 (5×5, no-pad)	5 (3×3, same-pad)
Channels	3→6→16→32→64	3→32→64→128→256→256
Spatial head	Flatten (6 400-d)	Glob. Avg Pool (256-d)
FC head	6400→120→84→1	256→128→1
Conv Bias	Yes	No (BN absorbs)
Optimiser	SGD	AdamW (wd=5e-3)
LR schedule	StepLR ($\gamma=0.1$)	CosineAnnealingLR
Sampler	Uniform	Weighted (class)
Public AUC	0.570	0.766

Figure 1: EfficientNet-B0 Stage1 training curves (**second_attempt_tuned**). Train AUC reaches 0.957 at the best val checkpoint (★, epoch 14) while val AUC peaks at 0.779, revealing a gap of 0.178. This overfitting motivated label smoothing, progressive unfreezing, and TTA.



References

- [1] Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *Proc. ICML 2019*.
- [2] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proc. CVPR 2016*, 770–778.
- [3] Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *Proc. ICLR 2019*.
- [4] Müller, R., Kornblith, S., & Hinton, G. (2019). When does label smoothing help? *Proc. NeurIPS 2019*, 4694–4703.
- [5] Graham, B. (2015). Kaggle diabetic retinopathy detection competition report. *University of Warwick*.